Behavioural Biometrics

* Signature & keystroke scan

Dynamic feature Capture

* Speed
* pressure
* stroke order
* pen lift pattern
* Acceleration
* Direction & angle
* Timing

Real time eg of : bank counter s/m pipeline :-

Sign in using digital signature :- (Rythm check)
Capture the dynamic (speed, pressure, stroke order etc are recorded by sensors).
↓
Template. (Built live template Dynamic data is converted into template)
↓
compare to find live template is similar to digital one.
↓
Approve or flag.

Strength

- High user acceptance
- Fits actually into existing document work flow
- Too hard to forge

Weak

- Need digitised hardware
- skilled forge to study screen
- Natural variability

keystroke

Features
- Unique typing pattern.

Components
- I/p capture device (Std keyboard, virtual keyboard)

Timing measurement engine (Background software record millisecond recorded by press/release

events)

- Feature extracting module (calculate dwell time / flight time, digraph latency).
- Template database (Each user reference typing profile built during enrollment)
- Decision

Key Stroke Verification

→ Static verification (password verification)

→ Continuous

Strength

- Non-intrusive
- Invisible
- Enable Continuous authentication throughout session.
- Hard to replicate randomly.

Weakness

- Rythm changes with mode
- Multiple enrollment samples required.

* What is measured?
* Hardware needed
* Posture case
* Deployment port
* Continuous monitoring

5 risk category

- Centralized storage vulnerability
  → Most immediate & risk category

Features

→ Mass exposure in breaches
→ Can't revoke compromised templates
→ Single point of failure.

* Function to creep.

events)

- Feature extracting module (calculate dwell time / flight time, digraph latency).
- Template database (Each user reference typing profile built during enrollment).
- Decision

Key Stroke Verification

→ Static verification (password verification)

→ Continuous

Strength	Weakness
• Non-intrusive	• Rythm changes with mode
• Invisible	• Multiple enrollment samples required.
• Enable Continuous authentication throughout session.	
• Hard to replicate randomly.	

* What is measures?
* Hardware needed
* Best use case
* Deployment post
* Continuous monitoring

5 risk category

- Centralized storage vulnerability
  → Most immediate & risk category

Features

→ Mass exposure in breaches
→ Can't revoke compromised templates
→ Single point of failure.

* Function to creep.

* Surveillance & Tracking
* Lack of Transparency & control
* Bias & Discrimination

where does privacy breaches occur

1. At sensor level
2. During Transmission
3. In storage
4. At the matches
5. Through metadata

privacy Vs Security

Biometric Authentication Security
Biometric privacy protection

International Standards Overview

ISO/IEC Biometric Standards

Standards	Focus Area	privacy Reference
ISO/IEC 19794	Biometric Data interchange format	Template structure Data element.
ISO/IEC 18795	Biometric performance Testing	Evaluation Methodologies.
ISO/IEC 24745	~~Bio API~~ Biometric Information protection	Template protection Requirements
ISO/IEC 24713	Bio API	Interface Specification
ISO/IEC 30136	Attack testing for template protection	Security Evaluation

22/9/2026

key biometric ISO/IEC 24745 → Biometric Information protection

Requirements

1. Renewabability (Revocability)
2. Unlinkability
3. Irreversability
4. performance preservation

Unlinkability

- Can not determine if 2 template is of 2 different persons
- privacy of multi s/w

Irreversability

- can not construct original biometric from protected template
- computatn is infeasible requirement to protect
- Template inversation attack

performance preservation

- protection should not significantly affect
- practical usability
- Balance b/w privacy & functionality

Template protection Technique

→ ISO/IEC 24745 → Biometric Information protection

Technical Approaches:

① Cancellable Biometrics

② Biometric Encryption / Fuzzy Extractors

③ Homomorphic Encryption

④ Secure Enclaves / Trusted Execution

→ Cancellable Biometrics

- Apply non involvable biometric features.
- Diff. parameters are applied to diff protected modality template
- Enables unlinkability

→ Biometric Encryption

- Fuzzy extractors are used
- Cryptographic keys are been generated.
- Strong irreversability

→ Homomorphic Encryption

- Not checking of decryption
- Emerging technology.
- Used in apples
- More secured one.

Regulatory Framework - GDPR (European Union)

Article - 9

- processing prohibited access.

Exception

* Explicit consent
* Social security.
* Substantial

Key Requirements

- Data protection impact assessment
- purpose limitation
- Data Minimization
- storage limitation

- ~~write to~~
- Right to erasor (to forgot)
- Reach notification within 72 hrs.

Regulatory Framework – BIPA (Illinois USA)

Biometric Information privacy Act.

- This is for Biometric ~~Data~~ Information & identification
- Retina, Iris, finger print → Identification

Requirements

- Written policy
- Informed concern.
- purpose limitation
- Resonable care
- Retention schedule

Regulatory Framework – India's DPDP Act

Digital personal Data protection – DPDP Act 2023

Key provitions

- ~~Concent~~ consent
- Notice
- purpose limitation
- Data minimization
- Storage limitation
- Security Safeguards
- Breadge Notification

principle Rights:

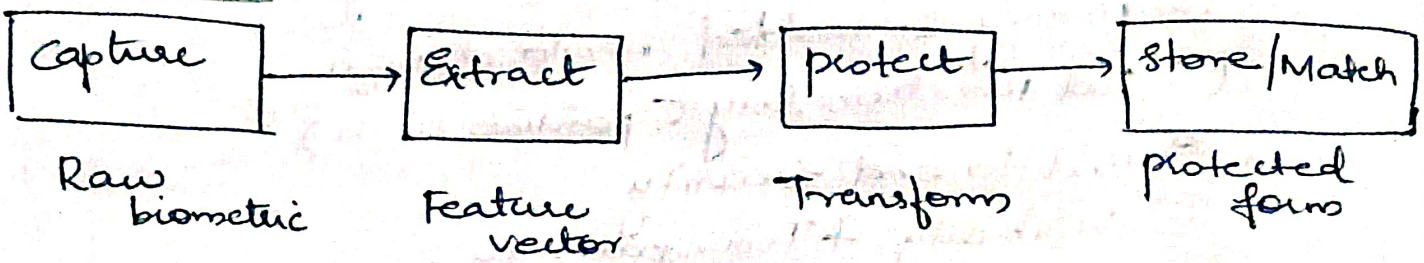
Right to Access
Right to Corrections / Erraswre
Right to revent redressal
Right to Nominate

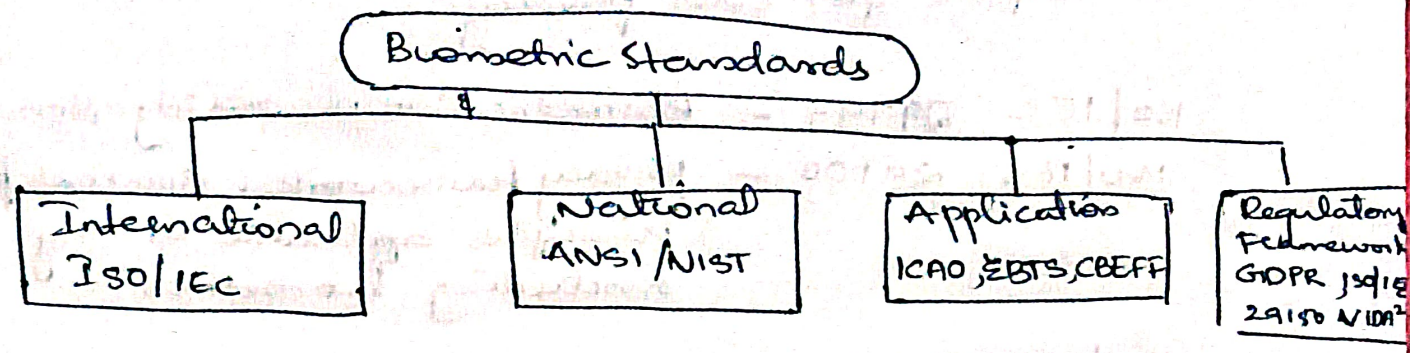
Aggenda

1. On-device processing
2. Decentralized storage
3. Zero knowledge proofs
4. Differential privacy.
5. Multiparty computation

1 → On device processing

24/9/2024 Pipeline





privacy sympathetic Biometric systems

- privacy protection is taking place.
- Reliable identification can be done.
- Cannot be reused.

permanance – It cannot revoked or ~~reused~~ reissued

Uniqueness – it enables cross linking of persons ~~identity~~ activity across unrelated s/ms. (function creep).

Involuntary disclosure – (faces/gaite, voice) can be taken without taking permission of the person.

sensitivity : It can reveal the health conditions, ethnicity and emotional state.

Irrevocability : Template compromise can lead to life threat of a person.

principles for designing privacy sympathetic Biometric s/ms

(Cavoukian's 7 foundational principles)

1. proactive, not reactive
2. privacy as the default setting
3. privacy Embedded into design
4. Full functionality (positive sum) (security + privacy are achieved together)
5. End to end security
6. Visibility & Transparency
7. Respect for user privacy.

ISO/IEC 29794 – Biometric sample quality standards

ISO/IEC 29100 – privacy framework. General privacy principles applicable to any IT s/ms including biometrics.

ISO/IEC 30107 – presentation attack detection (PAD) or liveliness detection called antispoofing.

ISO/IEC 24761 – Authentication context for biometrics.

ANSI/NIST ITL 1-2011 – Data format for the interchange of fingerprint facial & other biometric information.

NIST/FIPS 201 (PIV) – personal identity verification (credentials for US employees)

Application specific

ICAO Doc 9303 – Biometric e-Passports

FBI EBTs Electronic biometric transmission specification

CBEFF common biometric exchange formats framework.
(structure std for ISO/IEC 19785)